

OnlineSales.ai: Empowering global businesses with retail media solutions on the cloud

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ABOUT OnlineSales.ai



OnlineSales.ai

OnlineSales.ai launched its global retail media expansion plan, cutting server response time by 30%, and reducing processing query time by 30% with Google Cloud solutions.

About OnlineSales.ai

OnlineSales.ai (<https://www.onlinesales.ai/>) is an AI-driven platform that empowers retailers and marketplaces to tap into the tremendous potential of retail media, a rising trend that connects consumers with targeted online ads. The startup offers white-labeled solutions that are customized to each client's unique business needs. The platform also enables multi-outlet brands to efficiently boost marketing across sales partner networks.

Industries: Technology

Google Cloud results

- Enabled faster AI analytics for targeted advertising by cutting data warehouse query time by more than 60%
- Reduced ad placement time from 80 milliseconds to 55 milliseconds with support of Memystore and Cloud Load Balancing
- Freed engineers to focus on solution development by using Dataproc to cut two full

Slash
serve
time

Location: India

Load Balancing



About Niveus Solutions

Niveus Solutions is an award-winning Google Cloud partner that drives innovative cloud engineering solutions across India and Singapore. In 2020, it was named Asia-Pacific Breakthrough Google Cloud Partner of the Year.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Memystore
(<https://cloud.google.com/memystore/>)

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days of
provisioning
tasks

- Supported
customer
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complex
cloud
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through
seamless
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with zero
downtime

Retail media is a fast-growing field in which retailers offer advertising space on their websites, apps or marketplaces, leveraging consumer data to connect potential buyers to relevant ads in real-time.

According to Boston Consulting Group, the retail media market is expected to grow by 25% per year (<https://www.bcg.com/publications/2022/how-media-is-shaping-retail>)

to \$100 billion by 2027.

Amid such massive potential, Indian tech entrepreneurs Ashish Mehta and Kaushik Paranjape realized that many retailers lacked the artificial intelligence (AI) and data analytics know-how to capitalize on the opportunities through their own retail media solutions. The insight led them to found **OnlineSales.ai** (<https://www.onlinesales.ai/>), a plug-and-play whitelabel platform for retailers to leverage the

startup's AI algorithms, and display targeted banners in real-time as customers browse their sites.

OnlineSales.ai began life in 2017 with a computing architecture built on a leading global public cloud, using the provider's Kubernetes solution for scalable computing, and its data warehouse to drive AI data analytics. In early 2022, OnlineSales.ai made a bold move to migrate to [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>), even though it had years of familiarity with its previous provider. The decision came after it began testing the data processing speeds of BigQuery against those of the legacy data warehouse, and found an impressive improvement.

In addition to BigQuery advantages, OnlineSales.ai was also inspired to make the leap to Google Cloud thanks to lower latencies enabled by [Cloud Load Balancing](https://cloud.google.com/load-balancing) (<https://cloud.google.com/load-balancing>), and more seamless autoscaling with [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) (GKE). A final key factor, shares Kaushik Paranjape, CTO of OnlineSales.ai, was the amazing support from the Google Cloud team in architecting a comprehensive migration.

"We were initially a bit nervous about switching to Google Cloud after many years of familiarity with our previous cloud provider," says Paranjape. "When we discovered just how much BigQuery and Cloud Load Balancing could improve our ad server times, those doubts evaporated. The results have exceeded our expectations."

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Transforming ad server response time with BigQuery and Cloud Load Balancing

Complex cloud migration is a daunting prospect for any enterprise, let alone one that has been on the same provider for years. OnlineSales.ai was surprised to find the transition to be simple and seamless thanks to user-friendly Google Cloud solutions and a proactive Google Cloud support network.

With the support of partner [Niveus Solutions](https://niveussolutions.com/) (<https://niveussolutions.com/>) (Niveus), the retail media pioneer executed a full-fledged migration within four months, with zero downtime throughout the entire process. Niveus supported OnlineSales.ai in envisioning their architecture on Google Cloud, and eventually with the migration. The teams worked together to understand the various services and Niveus helped choose the best combination for the business.

After a painless transition, the business impact was felt immediately, shares Paranjape. In the retail media world, every second counts in the effort to attract consumer interest to online product offerings, amid competing demands for attention on an online marketplace. By adopting BigQuery as its core solution to power AI data analytics, OnlineSales.ai was able to slash its average data query time from eight seconds on its original provider, to three seconds with BigQuery, while incurring zero extra data warehouse cost.



At the same time, the partnership with Niveus meant that even post-migration, OnlineSales.ai had a team they could work with for ongoing support. The Niveus team operated like an extension of their in-house team, providing ongoing consultation where required.

As BigQuery transformed the company's data processing speeds, Cloud Load Balancing significantly improved network connection latencies by automatically linking server requests to the nearest node in Google Cloud's global computing network. The overall result saw OnlineSales.ai reducing ad server time, the time required to place advertisements on a website or mobile app, from 80 milliseconds to 55 milliseconds, a significant advantage in a lightning-fast industry.

"The faster shoppers are seeing ads on our customer websites, the more engaged they will be, and that leads to better sales. This ability is driven

by the amazing response times of BigQuery and Cloud Load Balancing," says Paranjape. "A big bonus is that while achieving better performance, we've also been able to save on overall computing costs compared to our legacy provider."

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Deploying Dataproc to free up teams from manual infrastructure tasks

OnlineSales.ai uses Apache Storm, an open-source data stream processing framework, to build powerful data pipelines into BigQuery for AI data analytics. Previously, the startup manually provisioned the Apache Spark infrastructure, a major burden on the DevOps team especially during peak sales periods such as Diwali or Black Friday.

Using Dataproc (<https://cloud.google.com/dataproc/>), a fully managed and scalable service for running Apache and other data solutions, Onlinesale.ai automates the provisioning of Apache Storm stream processing capabilities. That saves the engineering team two full days of manual provisioning chores during peak periods, a full day for scaling up to prepare for the surge, and another for scaling down. This in turn frees up the team to focus on creative app development tasks instead of mundane infrastructure.



Driving a global expansion strategy with advanced machine learning and microservices

Looking to the future, OnlineSales.ai is forging plans to take its retail media solution beyond its core Asian market to the United States, Europe and beyond. Google Cloud has been proactively assisting with the expansion through go-to-market strategies that leverage Google's global online advertising presence, in particular by matching the startup with prospective retailers.

On the product side, OnlineSales.ai is evaluating the adoption of BigQuery ML

(<https://cloud.google.com/bigquery-ml/docs/introduction>)

to take its machine learning capabilities to an even higher level, for greater speed and accuracy in enabling pinpointed banner displays. The startup is also considering the adoption of TensorFlow

(<https://cloud.google.com/tensorflow-enterprise>) and

Speech-to-Text

(<https://cloud.google.com/speech-to-text/>) to enhance AI capabilities, especially for improving user experience on customer websites.

As OnlineSales.ai enters new global markets, it anticipates massive surges in consumer traffic from around the world that will require increasingly precise and powerful autoscaling capabilities. Paranjape therefore plans to work with the Google

Cloud support team to develop more complex Kubernetes applications, which he says will become ever more critical as the company gains a wider global profile.

"We're now on a journey of rapid global expansion," says Paranjape. "Advanced Google Cloud solutions such as BigQuery ML and GKE will be key to enabling us to scale and develop more sophisticated AI algorithms, as we empower a fast-growing clientele eager to tap the unlimited potential of retail media."

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